Rojin Sharma

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Education

Vanderbilt University | Nashville, TN **Expected May 2027**

*Bachelor of Science in Computer Science, Mathematics & Physics (**Triple Major**)*

- **Relevant Coursework:** Artificial Intelligence, Data Structures, Algorithms, Software Design, Discrete Math, Computer Architecture, Probability & Statistics, Linear Algebra, Calculus
- **Awards:** International Olympiad on Astronomy and Astrophysics(**IOAA**), National Delegate**; Gold Medalist**, MILSET Expo Sciences**;** James C. Seus Scholarship(**full-ride**); Dean's List(**2024-present**)

Experience

Software Engineering Intern **Feb 2025 – Present**

A5 IT *Chicago, IL*

- Stabilized a failing backend service by refactoring its core architecture with an asynchronous job queue, cutting server load by **30%** and preventing recurring outages that risked client operations
- Overhauled security by integrating Auth0 (Okta) with granular RBAC, shutting down unauthorized access risks and earned recognition from leadership for eliminating a critical compliance gap
- Implemented strategic indexing and SQL optimization boosted database throughput by **40%**, removing a scaling bottleneck that blocked new client on-boarding

Software Engineering Intern **May 2024 – Aug 2024**

JumboTaxes(Automated Property Tax Appeal) *Philadelphia, PA*

- Upgraded the company's data pipeline by migrating a sluggish Django backend to Node.js, slashing document processing time by **35%** and unblocking revenue-critical automation workflows
- Engineered a hybrid PostgreSQL and MongoDB database solution, improving analytics query performance by **40%**
- Engineered robust error-handling and validation mechanisms in REST APIs, reducing failed submissions by **45%** and improving customer experience

Personal Projects

StudyBee | ***Retreival Agumented Generation(RAG), OpenAI, Multer, JWT , Pinecone , PostgreSQL, Vercel***

- Solo-launched v1 in 6 wks: Built **5 modules, 30+ REST APIs,** full cycle (**design→deploy→iterate**)
- Engineered automated document-processing pipeline: Parsed and vector-indexed **100 %** of course materials (PDF/Docs/Slides); delivered instant AI-driven student support course assistant **(<2 s, P95 latency)**
- Tackled fragmented file storage by implementing AWS S3 with IAM-based access, building a secure and scalable upload pipeline from the ground up
- Implemented GitHub Actions → Vercel CI/CD: Achieved **<5** min deployments with **0 downtime** on every push

RecallRay | ***Model Context Protocol(MCP), Agentic Workflow, RAG, Auth.js, MongoDB, OpenAI***

- Led 3-person team: Shipped beta **MVP in 4 weeks**; deployed live to **20+** beta users for context retrieval from Slack, Gmail, Zoom & Docs
- Indexed & processed **3M+** messages/documents: Built automated ingestion pipeline for seamless syncing, semantic indexing, and unified context retrieval across diverse data sources
- Integrated MCP enabling Instant semantic context retrieval from Google Drive & Docs **(<350 ms latency, P95)**

Harmony Quest – Guitar Sound Simulator | — *C++*

- Built a guitar sound synthesis engine using the Karplus–Strong algorithm for realistic tone generation.
- Optimized the engine for low-latency playback and an extensible library of **37 notes spanning 3 octaves**

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C++, Java
Frameworks & Technologies: React, Node.js, Express.js, Next.js, HTML, Tailwind CSS, OpenAI, Postman
Tools and Systems: Git (Version Control), GitHub Actions, REST APIs, CI/CD, Docker, Vercel, AWS(S3), Agile, PostgreSQL(SQL), TypeORM, MongoDB(NoSQL), Pinecone(Vector Database), Model Context Protocol(MCP), Pandas, NumPy, MatplotLib, Scikit-learn.